Practical 1

Aim: Design a Web Application for an Organization with Registration forms and advanced controls.

Theory

1. Label

A label is a user for placing text inside the container. A label is used only for inputting text. The label does not imply that the text can be altered or it can be used as a button which can be further worked upon.

2. Button

This command generates a button in the User Interface. Clicking on the button would move the command to another page or another web server which is used to show several other outputs in the user interface page.

3. Checkbox

There can be a certain question and the checkbox is used to determine the true or false nature of the question being asked. If the checkbox is ticked then it means that the said question is true which if it is unchecked it means that the said question is false. It is basically a true or false state in Java programming language.

4. Checkbox Group

As the name implies the checkbox group is a set of checkboxes that are being used in the programming language. There are many checkboxes that are being used and hence the group of checkboxes is known as the checkbox group.

5. List

The list gives a scrolling list of items for the user. The scrolling list of items is also being set by the user. The user sets the scrolling list of items such as Fruits, Vegetables, some questionnaire, or other facts.

6. Text Field

A text field is used for the editing of a particular line of text which can be used within the programming concept.

7. Text Area

A text area is used for the editing of multiple lines of text. The only difference between the Text field and Text area is that Text Field is used for editing a single line of text within the user interface while a Text Area is used for editing multiple lines of text.

8. Choice

A choice, as the name implies, shows the various options and the choice that is selected is shown in the top menu bar of the screen.

9. Canvas

In the canvas space, there can be an input being given by the user or the user can draw something on the Canvas space being given.

10. Image

There can be a single image or multiple images within a UI. There can be a button being associated with an image and when it is clicked it can produce some functionality.

11. Scroll Bar

The scroll bar like a normal one is used to scroll or move from a varied range of values. The user selects one value from those range of values.

12. Dialog

The dialog is used to take some form of input from the user and produce it in a sequential manner.

13. File Dialog

From a file dialog, a user can select a file which he/she wishes to use.

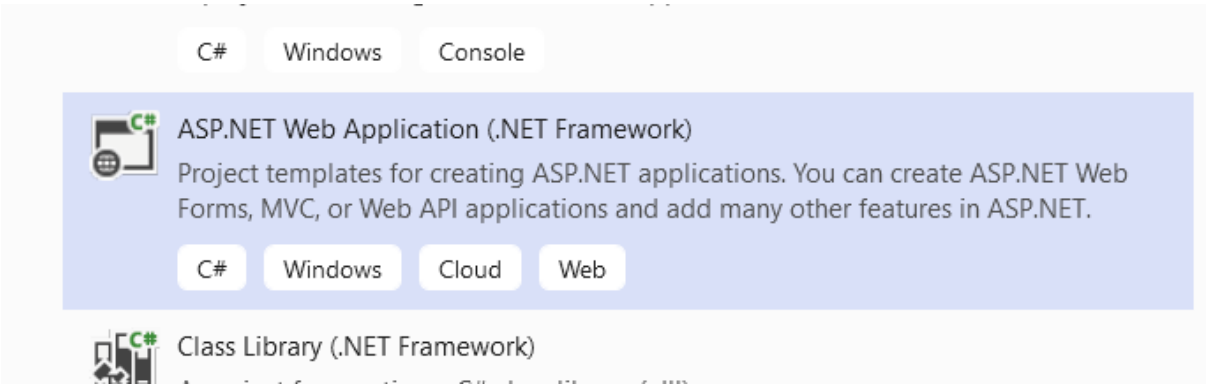
STEP 1. Open Microsoft Visual **Studio**

Click **Create a new project**

STEP 2: Create ASP.NET Web Application

1. Select **ASP.NET Web Application (.NET Framework)**

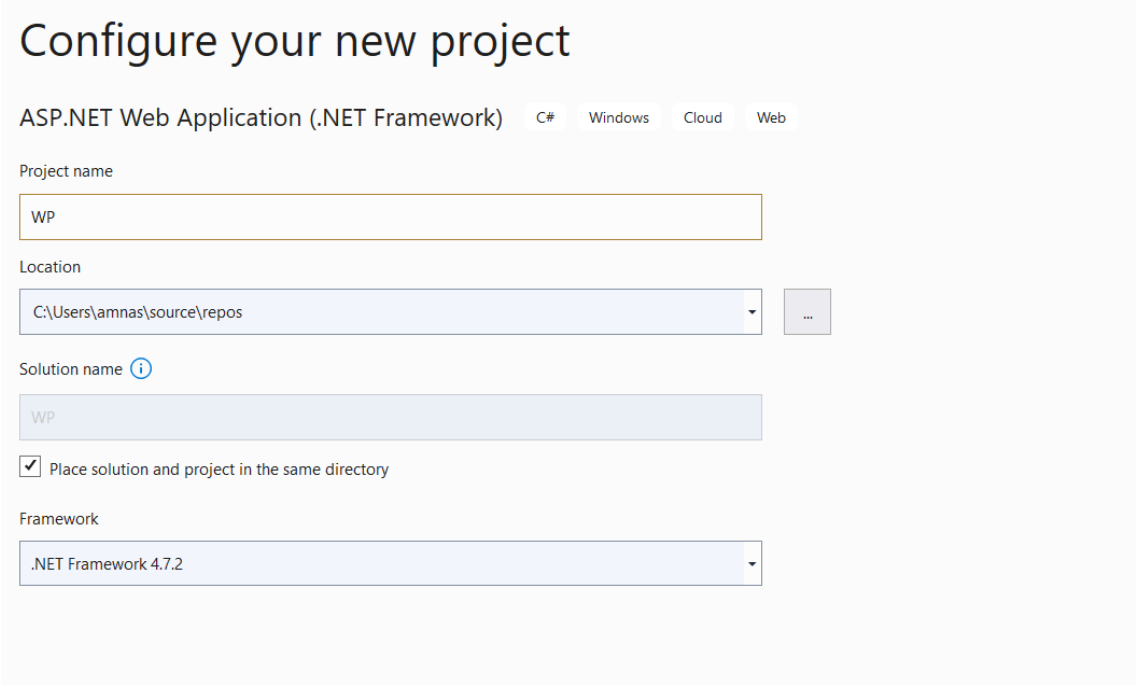
2. Click **Next**



3. Project Name:

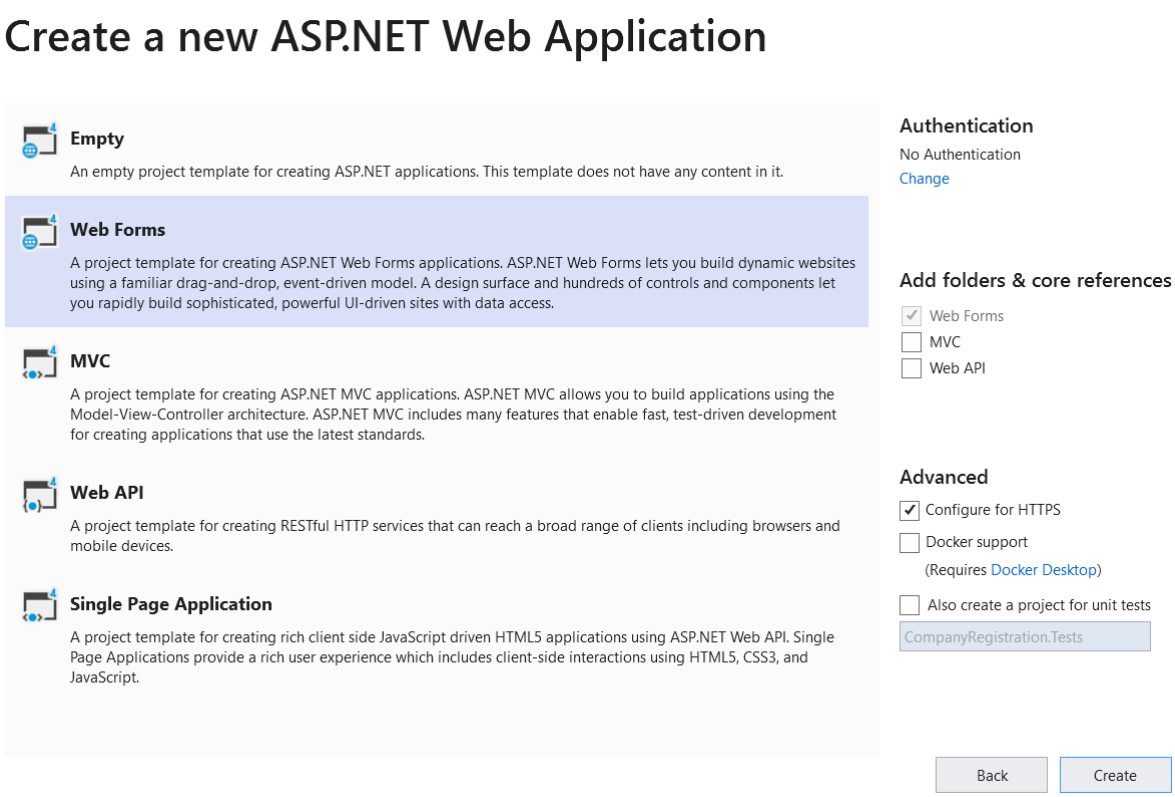
   WP

4. Choose Location → Click **Create**



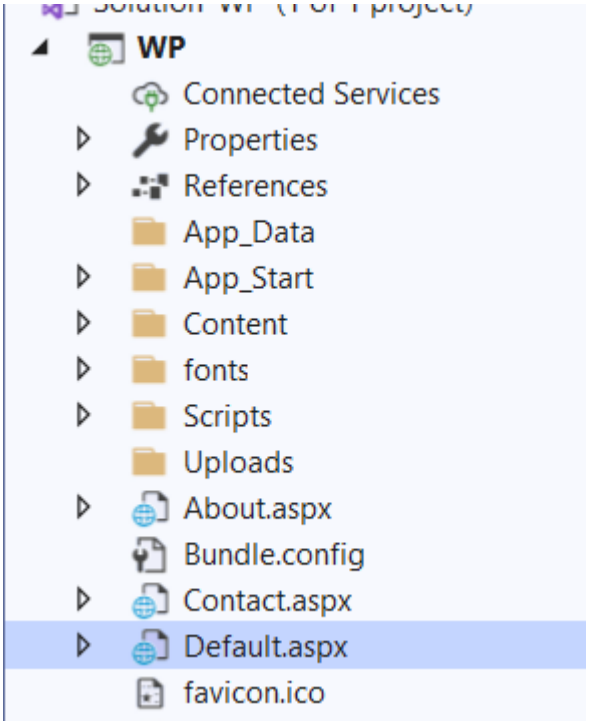
STEP 3: Choose Project Template

1. Select **Web Forms**
2. Authentication: **No Authentication**
3. Click **Create**



Visual Studio creates:

- `Default.aspx`
- `Default.aspx.cs`
- `Web.config`



Code:

Default.aspx:

```
<%@ Page Title="Home Page" Language="C#" AutoEventWireup="true"
CodeBehind="Default.aspx.cs" Inherits="WP._Default" %>

<!Doctype html>
<html>
<head runat="server">
    <Title> Registration Form </Title>
    <style type="text/css">
        .auto-style1 {
            width: 100%;
        }
        .auto-style2 {
            width: 186px;
        }
        .auto-style4 {
            height: 32px;
        }
    </style>
</head>

<body>
<form id="form1" runat="server">
    <table class="auto-style1">
        <tr>
            <td colspan="2" style="text-align: justify; font-family: 'Times New
Roman', Times, serif; font-size: 26px; font-weight: bold; font-style:
normal;">Registration Form</td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label1" runat="server" Text="Name"></asp:Label>
            </td>
            <td>
```

```
                <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label2" runat="server" Text="Email"></asp:Label>
            </td>
            <td>
                <asp:TextBox ID="txtEmail" runat="server"
TextMode="Email"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label3" runat="server" Text="Password"></asp:Label>
            </td>
            <td>
                <asp:TextBox ID="txtPassword" runat="server"
TextMode="Password"></asp:TextBox>
                <asp:RequiredFieldValidator ID="rfvPassword" runat="server"
ErrorMessage="Password Required "
ControlToValidate="txtPassword"></asp:RequiredFieldValidator>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label4" runat="server" Text="Confirm
Password"></asp:Label>
            </td>
            <td>
                <asp:TextBox ID="txtConfirmPassword" runat="server"
TextMode="Password"></asp:TextBox>
                <asp:CompareValidator ID="cvConfirmPassword" runat="server"
ControlToCompare="txtPassword" ControlToValidate="txtConfirmPassword"
ErrorMessage="Password doesn't match"></asp:CompareValidator>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label5" runat="server" Text="Gender"></asp:Label>
            </td>
            <td>
                <asp:RadioButton ID="rdoMale" runat="server" Text="Male" />
 <asp:RadioButton ID="rdoFemale" runat="server" Text="Female" />
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label6" runat="server" Text="Course"></asp:Label>
            </td>
            <td>
                <asp:CheckBox ID="chkJava" runat="server" Text="Java" />
 <asp:CheckBox ID="chkPython" runat="server" Text="Python" />
 <asp:CheckBox ID="chkASP" runat="server" Text="ASP.Net" />
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label7" runat="server" Text="Department"></asp:Label>
            </td>
            <td>
                <asp:DropDownList ID="ddlDepartment" runat="server">
```

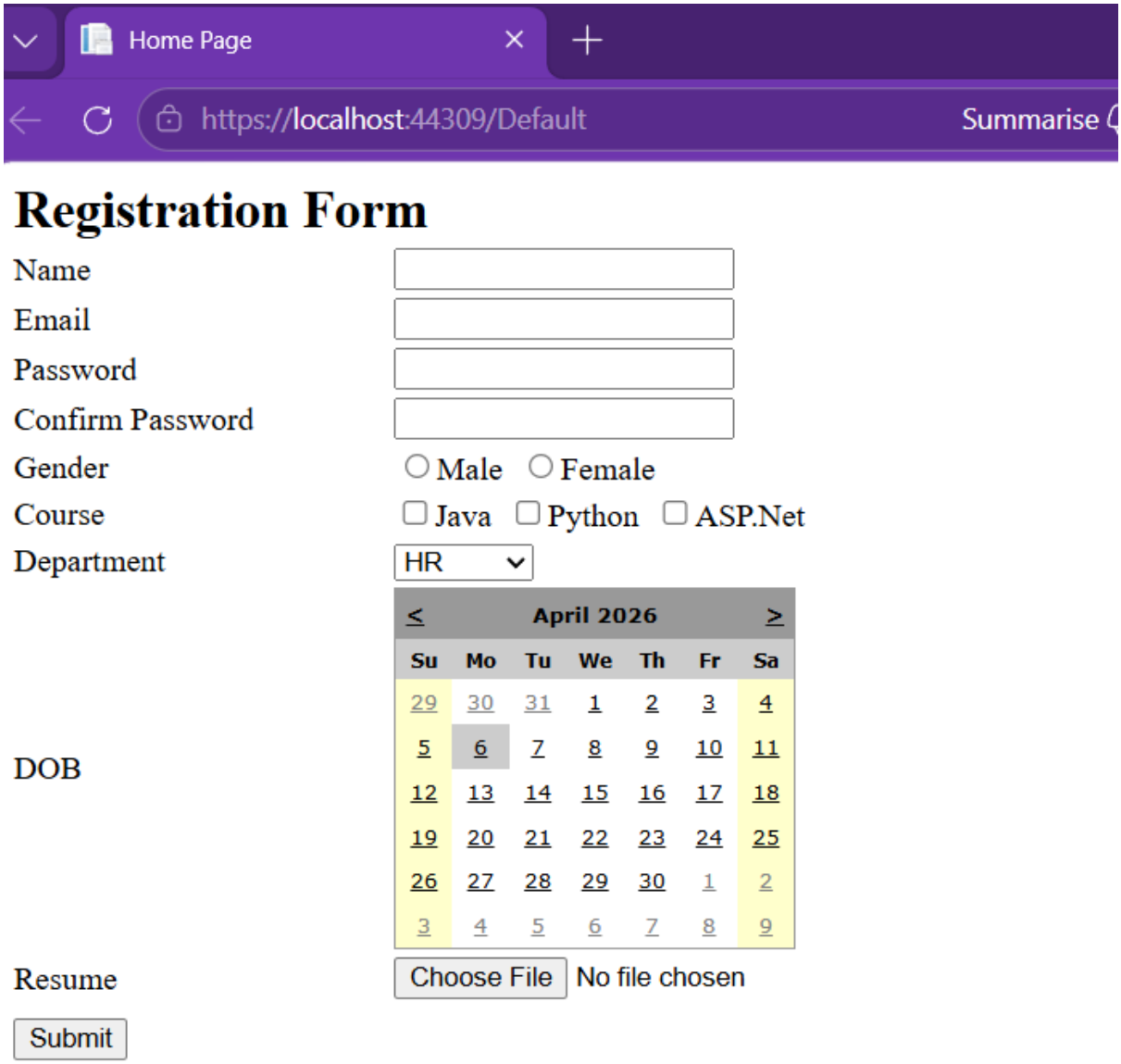
```
                    <asp:ListItem>HR</asp:ListItem>
                    <asp:ListItem>IT</asp:ListItem>
                    <asp:ListItem>Finance</asp:ListItem>
                </asp:DropDownList>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label8" runat="server" Text="DOB"></asp:Label>
            </td>
            <td>
                <asp:Calendar ID="calDOB" runat="server" BackColor="White"
BorderColor="#999999" CellPadding="4" DayNameFormat="Shortest" Font-Names="Verdana"
Font-Size="8pt" ForeColor="Black" Height="180px" Width="200px">
                    <DayHeaderStyle BackColor="#CCCCCC" Font-Bold="True" Font-
Size="7pt" />
                    <NextPrevStyle VerticalAlign="Bottom" />
                    <OtherMonthDayStyle ForeColor="#808080" />
                    <SelectedDayStyle BackColor="#666666" Font-Bold="True"
ForeColor="White" />
                    <SelectorStyle BackColor="#CCCCCC" />
                    <TitleStyle BackColor="#999999" BorderColor="Black" Font-
Bold="True" />
                    <TodayDayStyle BackColor="#CCCCCC" ForeColor="Black" />
                    <WeekendDayStyle BackColor="#FFFFCC" />
                </asp:Calendar>
            </td>
        </tr>
        <tr>
            <td class="auto-style2">
                <asp:Label ID="Label9" runat="server" Text="Resume"></asp:Label>
            </td>
            <td>
                <asp:FileUpload ID="fileUpload" runat="server" />
            </td>
        </tr>
        <tr>
            <td class="auto-style4" colspan="2" style="text-align: justify">
                <asp:Button ID="btnSubmit" runat="server" Text="Submit"
OnClick="btnSubmit_Click" />
            </td>
        </tr>
        <tr>
            <td colspan="2" style="text-align: justify">
                <asp:Label ID="lblMessage" runat="server"
ForeColor="#00CC66"></asp:Label>
            </td>
        </tr>
        <tr>
            <td class="auto-style2"> </td>
            <td> </td>
        </tr>
    </table>
</form>
</body>
</html>
```

Default.aspx.cs:

```
using System;
```

```
using System.IO;

namespace WP
{
    public partial class _Default : System.Web.UI.Page
    {


        // Button Click Event
        protected void btnSubmit_Click(object sender, EventArgs e)
        {
            // 1. Get TextBox Values
            string name = txtName.Text;
            string email = txtEmail.Text;
            string password = txtPassword.Text;

            // 2. Gender (RadioButton)
            string gender = "";
            if (rdoMale.Checked)
                gender = "Male";
            else if (rdoFemale.Checked)
                gender = "Female";

            // 3. Course (CheckBox - Multiple)
            string course = "";

            if (chkJava.Checked)
                course += "Java ";

            if (chkPython.Checked)
                course += "Python ";

            if (chkASP.Checked)
                course += "ASP.NET ";

            // 4. Department (Dropdown)
            string department = ddlDepartment.SelectedValue;

            // 5. Date of Birth (Calendar)
            string dob = "";
            if (calDOB.SelectedDate != DateTime.MinValue)
                dob = calDOB.SelectedDate.ToShortDateString();
            else
                dob = "Not Selected";

            // 6. File Upload
            string fileName = "";

            if (fileUpload.HasFile)
            {
                fileName = fileUpload.FileName;

                // Create path
                string path = Server.MapPath("~/Uploads/") + fileName;

                // Save file
                fileUpload.SaveAs(path);
            }
            else
            {
                fileName = "No file uploaded";
            }
```

```
            // 7. Display Output
            lblMessage.Text = "<b>Registration Successful!</b><br/><br/>" +
                              "Name: " + name + "<br/>" +
                              "Email: " + email + "<br/>" +
                              "Gender: " + gender + "<br/>" +
                              "Course: " + course + "<br/>" +
                              "Department: " + department + "<br/>" +
                              "DOB: " + dob + "<br/>" +
                              "File: " + fileName;
        }
    }
}
```



If you are Selecting Empty Web Form while starting the new project you might get JQuery error:

Server Error in '/' Application.

WebForms UnobtrusiveValidationMode requires a ScriptResourceMapping for 'jquery'. Please add a ScriptResourceMapping named jquery(case-sensitive).

Description: An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more information about the error and where it originated in the code.

Exception Details: System.InvalidOperationException: WebForms UnobtrusiveValidationMode requires a ScriptResourceMapping for 'jquery'. Please add a ScriptResourceMapping named jquery(case-sensitive).

Source Error:

An unhandled exception was generated during the execution of the current web request. Information regarding the origin and location of the exception can be identified using the exception stack trace below.

Stack Trace:

```
[InvalidOperationException: WebForms UnobtrusiveValidationMode requires a ScriptResourceMapping for 'jquery'. Please add a ScriptResourceMapping named
   System.Web.UI.ClientScriptManager.EnsureJqueryRegistered() +143
   System.Web.UI.WebControls.BaseValidator.RegisterUnobtrusiveScript() +11
   System.Web.UI.WebControls.BaseValidator.OnPreRender(EventArgs e) +88
   System.Web.UI.Control.PreRenderRecursiveInternal() +166
   System.Web.UI.Control.PreRenderRecursiveInternal() +236
   System.Web.UI.Control.PreRenderRecursiveInternal() +236
   System.Web.UI.Page.ProcessRequestMain(Boolean includeStagesBeforeAsyncPoint, Boolean includeStagesAfterAsyncPoint) +4714
```

Version Information: Microsoft .NET Framework Version:4.0.30319; ASP.NET Version:4.8.9319.0

Since Modern ASP.NET validation needs jQuery library

In that case open Web.Config File and paste the below code:

```
<appSettings>
  <add key="ValidationSettings:UnobtrusiveValidationMode" value="None" />
</appSettings>
```

Web.config | Default.aspx.cs | Default.aspx | Registration Form: Overview

```
    <compilation debug="true" targetFramework="4.7.2" />
    <httpRuntime targetFramework="4.7.2" />
  </system.web>
  <system.codedom>
    <compilers>
      <compiler language="c#;cs;csharp" extension=".cs" type="Microsoft.CodeDom.P
      <compiler language="vb;vbs;visualbasic;vbscript" extension=".vb" type="Micr
    </compilers>
  </system.codedom>
    <appSettings>
        <add key="ValidationSettings:UnobtrusiveValidationMode" value="None" />
    </appSettings>
</configuration>
```

And then run the program again.